The Representational Regress: Machine-Checked Topology, Diagonal Arguments, and Boundary Obstructions for Representational Systems

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2026

Abstract

This work is a Lean 4 formalization, checked against Mathlib, with zero `sorry` and no project-specific axioms beyond the Lean core. It treats *representational systems*—categories equipped with a non-trivial iterated self-map—and an intrinsic planar-chart predicate `ChartableR2` on compact topological models.

(i) **Categorical strand.** A representational system yields infinitely many formally distinct meta-levels; Lawvere's fixed-point theorem in `Type` and in `MonoidalClosed(Type)` is complemented by an inductive `OntologicalSlot` witness that separates morphism data from global-element data, so the "Lawvere wall" does not collapse that distinction inside the formalism.

(ii) **Topological strand.** The closed cylinder $S^1 \times [0,1]$ and the Möbius strip (quotient of $[0,1]^2$ by $(0,t) \sim (1,1-t)$) are not homeomorphic. Points that admit an open neighborhood homeomorphic to $\mathbb{R}^2$ are classified by `ChartableR2`; on both models the chosen boundary agrees with "not `ChartableR2`," while the boundary *subspaces* differ (connected vs. disconnected). A general lemma, `not_homeomorphic_of_chartableR2_boundary_contrast`, states the same obstruction for any two spaces once those biconditionals and a subtype non-homeomorphism are supplied.

(iii) **Bridges to neighboring debates.** The formal library also includes `SymbolSystem` (synonymous with `RepresentationalSystem`) and lemmas restating regress injectivity in the vocabulary of interpretation; `LawvereRegressBridge`, which derives impossible "universal parametric" enumerations of unary maps and relates Lawvere phenomena to `OntologicalSlot`; bundled `ChartableR2BoundaryModel` instances for the cylinder and Möbius strip; and `QuantumObserverChainHypothesis`, a structure whose fields are exactly the premises of the general boundary lemma—intended for a *hypothetical* topological model of an observer–readout chain, without encoding Hilbert spaces or measurement axioms in the present development. No theorem refers to consciousness; a supplementary philosophical premise P2, that a target notion of self-presence is modeled globally by Möbius-type rather than cylinder-type organization, combines with (ii) to yield a conditional reading under which representational substrates matching the cylinder-side boundary pattern cannot be homeomorphic to a space matching the Möbius-side pattern—a gap that, when those premises apply, is not removed by scaling, iteration, or continuous deformation within a fixed topological type.

(iv) **No-escape packaging**
(`NoEscapeClosure`, `LawvereRegressBridge`).
A separate strand makes adversarial "horn" moves explicit, closing six routes by which an opponent might claim to circumvent the regress or the Lawvere wall:

(a) Iterate collisions force a *finite* repertoire of monoid powers—no infinite injective $\mathbb{N}$-tower along a periodic channel.
(b) A section–retraction readout induces an idempotent loop on the host and the identity on the source, blocking injective towers on *both* sides.
(c) Even *partial* (non-surjective) self-models necessarily miss their own diagonal—no surjectivity assumption required.
(d) A decidable/computability analogue closes the finite Lawvere horn using $\neg : \mathrm{Bool} \to \mathrm{Bool}$.

These lemmas do not replace `iter_injective`, but spell out what happens when a rival weakens, reroutes, or partially implements self-representation.

The source tree contains roughly twenty-five Lean modules; the body and appendices summarize the `OntologicalSlot` construction, seam and equator charts for `ChartableR2`, scope, and reproduction.

1 Introduction

1.1 The Problem

Representational systems—symbolic AI, formal theories, or any setting where a structure represents itself by iterating a fixed construction—support arbitrarily deep *meta*-levels. Classical diagonal and fixed-point arguments constrain how much "self-capture" such a layer can achieve. Independently, one may ask whether intrinsic topology distinguishes different *kinds* of "inside vs. outside" organization when global homeomorphism is taken as geometric sameness.

This paper gives a Lean 4 formalization of a sizeable fragment of that joint landscape. The technical development stays in categorical and topological language; §7.3 relates the results to consciousness, symbolic grounding, diagonal arguments, and a template for observer–apparatus topologies.

1.2 Summary of Results

The formalization proves, with zero `sorry` and no custom axioms, a **layered** constraint on representational systems. The layers are logically independent: one can accept the topological results without leaning on an infinite tower, and one can study the tower without the cylinder/Möbius contrast. Together they imply that "more representation" is not a generic path from one global geometric type to another.

(a) **Categorical regress and Lawvere wall** (§2): under a bundled non-collapse hypothesis on iterates of `represent`, there are infinitely many formally distinct meta-levels; Lawvere's theorem in `Type` / CCC form is complemented by `OntologicalSlot`, so a fixed-point tally cannot erase the morphism-vs-global-element typing.

(b) **Universal topological obstruction** (§4.3, §6): Theorem 4.4 states a *template*—if two spaces admit subsets characterized as exactly the non-`ChartableR2` points, and if those boundary *subspaces* are not homeomorphic, then the ambient spaces are not homeomorphic. **M-FINAL** is a fully worked compact witness (cylinder vs. Möbius strip). Any future domain whose state space can be modeled with the same sensor pattern and an incompatible boundary subtype imports the same conclusion; no separate bridge argument is required.

(c) **Extended master bundle** (§7): `representational_regress_master_extended` adjoins the 1D half-space non-homeomorphism and M-FINAL to the core master conjunction.

(d) **Cross-domain bridges** (§7.3):

- `SymbolSystem`
- `LawvereRegressBridge`
- concrete `ChartableR2BoundaryModel` instances
- `QuantumObserverChainHypothesis`

These make the obstruction's *hypothesis shape* explicit for grounding, diagonal, and observer–readout readings.

(e) **Adversarial horn closures** (§2, §7.6).
Six escape routes are formally blocked in `NoEscapeClosure` and `LawvereRegressBridge`: iterate collision dichotomy, both-sided section–retraction idempotent collapse, diagonal exclusion for partial self-models (no surjectivity required), decidable Lawvere for Bool-valued functions, and the original $\mathbb{N}$-valued enumerator obstruction. Together these leave an adversary with a clean dichotomy: either an infinite injective regress with Lawvere wall intact, or a finite loop too shallow for full self-capture.

Significance and what is *not* being claimed. Much informal discussion stops at *infinite regress*: every meta-representation invites another, so a desired "closure" fails. That phenomenon is real in the formalism (item (a)), but it is not the deepest punchline. The topological strand (item (b)) is **not** a complaint about non-termination: homeomorphism obstructions are *finite-data invariants* in spirit—two spaces are either homeomorphic or not, regardless of whether one enumerates infinitely many modeling steps. The barrier is that *no* continuous global identification (a homeomorphism) can exist between disparate types once the `ChartableR2`-boundary pattern and subtype mismatch are in place. In particular, the obstacle is not removed by "more compute," longer iterates, richer self-models, or architectural reparameterization if those changes do not alter the underlying topological type of the modeled substrate. Infinity can distract from the point; the point is **wrong type**, not **inexhaustible queue**.

Utility. For any application—symbolic AI, formal theories, measurement chains, or philosophical models—the workflow is: (1) encode the system (or its state space) in sufficient topological detail to verify the boundary biconditionals, (2) compare boundary subspaces up to homeomorphism, (3) invoke the general lemma or a bundled model. If the premises hold, the obstruction applies; if they fail, the obstacle was never licensed. That conditional structure is what keeps the mathematics honest while still delivering a *portable* gate.

1.3 Contributions and Novelty Claims

(1) `OntologicalSlot` — inductive tagging of morphism slots versus global-element slots; wall lemmas follow from constructor disjointness yet make the Lawvere-adjacent moral explicit inside the type discipline. [[Novel]]

(2) `ChartableR2` **spine + seam–equator charts** — intrinsic $\mathbb{R}^2$ chartability as a definable "planar interior vs. boundary" sensor; constructive charts through the quotient seam including the equator via piecewise unfolding. [[Novel] application / technique]

(3) **General boundary-contrast obstruction.**
`not_homeomorphic_of_chartableR2_boundary_contrast`
A single theorem that *packages* the logical move used in M-FINAL: any two spaces hit the

same wall once their `ChartableR2`-co-characterized boundaries are topologically incompatible. Individual proofs of non-homeomorphism are classical; a *reusable, machine-checked* lemma with this generality, wired to a concrete compact witness and to a categorical regress bundle, is—to our knowledge—*not* standard in the philosophy-of-mind or AI-safety literatures (where topology more often appears metaphorically). We claim distinctiveness for the **integrated, checkable** template, not for the elementary fact that the Möbius strip and the cylinder differ. [[Novel]]

(4) **Supplementary structures:**

- `SymbolSystem`
- `LawvereRegressBridge`
- bundled `ChartableR2BoundaryModel` data
- `QuantumObserverChainHypothesis`

These spell out which hypotheses must be imported from a domain before the obstruction applies; they do not expand the set of unconditional theorems but sharpen *where* the gate sits. [[Novel]]

(5) `NoEscapeClosure` + **extended** `LawvereRegressBridge`
Packages six complementary "escape routes," each formally blocked:

(i) Iterate collision ⇒ finite power repertoire.
(ii) Section–retraction readout ⇒ idempotent on host, identity on source; no injective tower on either side.
(iii) Partial (non-surjective) self-model ⇒ diagonal blind spot.
(iv) Decidable evaluator into Bool ⇒ no universal parametric unary.
(v) Full Lawvere for $\mathbb{N}$ and general B with fixed-point-free endo.
(vi) Wall persists at every iterate depth.

[[Novel]]

(6) **Classical ingredients.** Lawvere's diagonal argument (1969), textbook topology of the cylinder and Möbius strip, and Cantorian regress patterns precede this work. The contribution is a **unified, adversarially checked** landscape in which (i) regress and Lawvere infrastructure, (ii) a *general* topological obstruction, and (iii) a discharged witness and extension hooks coexist in one artifact, so that "compute harder" is not confusable with "change homeomorphism type."

1.4 Paper Organization

Section 2: representational systems, regress, Lawvere, `OntologicalSlot`, core master theorem. Section 3–6: cylinder and Möbius models, `ChartableR2`, generic lemma, seam charts, M-FINAL. Section 7: extended master and instantiations (consciousness, grounding, diagonal arguments, QM schematic). Sections 8–10: limits, related work, conclusion. Appendices: Lean module map, theorem index, reproduction.

2 The Representational Regress

2.1 The `RepresentationalSystem` Structure

Definition 2.1 (`Basic.lean`: `RepresentationalSystem`)**.**

A representational system *is a tuple* $(C, A, \text{represent}, \text{iter_injective})$ *where:*

- C *is a category,*
- A *is an object of* C*,*
- $\text{represent} : A \to A$ *is an endomorphism,*
- iter_injective*: for all* $n \neq m$*,* $\text{represent}^n \neq \text{represent}^m$ *in* $\text{End}(A)$*.*

2.2 Non-Termination and Unboundedness

Theorem 2.2 ([Known math] `Regress.lean`)**.** *For any representational system* R*:*

(a) *(`regress_no_termination`) Distinct indices* $n \neq m$ *yield distinct meta-level arrows.*

(b) *(`regress_iterates_unbounded`) For every bound* B*, there exists a level* $L > B$ *with a distinct iterate.*

(c) *(`regress_is_infinite`, `meta_range_infinite`) The set of meta-levels is infinite.*

Remark 2.3 (What if `iter_injective` fails?)**.** *The hypothesis* iter_injective *is* bundled*, not derivable from a bare endomorphism. An adversary might weaken it: suppose* $\text{represent}^n = \text{represent}^m$ *for some* $n \neq m$*.* `NoEscapeClosure` *formalizes the consequence: let* $e := \text{End.of}(\text{represent})$*; if* $e^{n+d} = e^n$ *with* $d > 0$ *then every power* e^k *equals some* e^i *with* $i < n + d$ *(`IterateClosure.End.exists_lt_of_pow_collision`).* `IterateClosure.End.pow_iterate_dichotomy` *records the* ***dichotomy****: for every monoid element* e*, either (i) distinct natural-number powers are all distinct (an infinite injective* $\mathbb{N}$*-tower), or (ii) the set of* distinct *powers is bounded by a finite index. In case (ii) the system has only finitely many representational layers—it loops. Either route blocks "just enough" self-representation: one gives an infinite regress; the other gives a finite loop too impoverished to host injective iterate depth.*

2.3 The Lawvere Wall

Theorem 2.4 ([Known math] `Lawvere.lean`)**.** *(`lawvere_fixed_point_Type`) If* $s : A \to (A \to B)$ *is surjective, then every* $f : B \to B$ *has a fixed point.*

Theorem 2.5 ([Known math] `LawvereCCCType.lean`)**.** *(`lawvere_fixed_point_MonoidalClosedType`) The same conclusion holds when the universal-enumerator hypothesis is recast as surjectivity of* $\text{curry}(\text{lawvereBinary } s)$ *in* $\text{MonoidalClosed}(\text{Type } u)$*.*

2.4 The `OntologicalSlot` Construction

Definition 2.6 ([Novel] `Basic.lean`: `OntologicalSlot`)**.** *For a category* C *with terminal object and an object* A*:*

$$\text{OntologicalSlot}(C, A) \;:=\; \text{endo}(f : A \to A) \;\mid\; \text{obj}(g : \top_C \to A)$$

an inductive type with two constructors, disjoint by definition.

Theorem 2.7 ([Novel] `FixedPoints.lean`)**.** *(`lawvere_wall_is_not_dissolution`) For any representational system* R *and any global element* $\text{fp} : \top_C \to A$*,*

$$\text{OntologicalSlot.endo}(R.\text{represent}) \;\neq\; \text{OntologicalSlot.obj}(\text{fp}).$$

2.5 The Core Master Theorem

Theorem 2.8 (`Main.lean`). *(`representational_regress_master`) For any representational system R:*

$$\text{representational_regress_master_claim}(R)$$

i.e., the conjunction of unbounded regress, Lawvere wall non-dissolution, and T_2 invariance under homeomorphism.

3 Topological Models: Cylinder and Möbius Strip

3.1 The Closed Cylinder

Definition 3.1 (`CylinderBoundary.lean`). $\text{ClosedCylinder} := S^1 \times [0,1]$ *with the product topology and manifold structure* $\mathfrak{R}_1 \times \mathfrak{R}_1^{\partial}$.

Theorem 3.2 ([Known math] `CylinderBoundary.lean`). *The manifold boundary of* ClosedCylinder *decomposes as:*

$$\partial(\text{ClosedCylinder}) = F_{\text{bot}} \sqcup F_{\text{top}}$$

where $F_{\text{bot}} = S^1 \times \{0\}$ and $F_{\text{top}} = S^1 \times \{1\}$ are each connected, and their union is disconnected.

3.2 The Möbius Strip

Definition 3.3 (`CylinderMobius.lean`). *The* Möbius equivalence $\sim$ *on* $[0,1]^2$ *is the equivalence relation generated by* $(0,t) \sim (1, 1-t)$ *for* $t \in [0,1]$, *extended by reflexivity, symmetry, and transitivity. The* Möbius strip *is* $\text{MobiusStrip} := [0,1]^2/{\sim}$.

Theorem 3.4 (`CylinderMobius.lean`).

(a) (`mobiusRel0Graph_isClosed`) The graph of $\sim$ is closed in $[0,1]^2 \times [0,1]^2$.

(b) (`CompactSpace MobiusStrip`) MobiusStrip *is compact (quotient of compact).*

(c) (`instT2SpaceMobiusStrip`) MobiusStrip *is Hausdorff (closed graph + compact domain).*

(d) (`isConnected_mobiusStripBoundary`) The boundary band $\partial_{\text{model}} := \pi(\{t = 0\} \cup \{t = 1\})$ is connected (one component).

3.3 Boundary Subspace Contrast

Theorem 3.5 (`MobiusCylinderBoundaryContrast.lean`).
(`mobiusStripBoundary_not_homeomorphic_closedCylinderBoundaryUnion`)

$$\text{IsEmpty}(\text{Subtype}(\partial_{\text{Mob}}) \simeq_t \text{Subtype}(\partial_{\text{Cyl}}))\,.$$

The Möbius boundary band (connected) and the cylinder boundary union (disconnected) are not homeomorphic as subspaces.

4 The `ChartableR2` Framework

4.1 Definition and Basic Properties

Definition 4.1 (`ChartableR2Neighbor.lean`). *A point x of a topological space X is* ChartableR2 *if there exists an open set $U \ni x$ such that $U \cong_t \mathbb{R}^2$.*

Theorem 4.2 (`ChartableR2Neighbor.lean`).

(a) (`isOpen_setOf_chartableR2`) The set of `ChartableR2` points is open.

(b) (`isClosed_setOf_not_chartableR2`) Its complement is closed.

(c) (`not_chartableR2_of_tendsto_seq_atTop`) Sequential stability: limits of non-chartable points are non-chartable.

4.2 Invariance Under Homeomorphism

Theorem 4.3 (`ChartableR2Bridge.lean`).
(`Homeomorph.chartableR2_iff`)
If $h : X \simeq_t Y$ is a homeomorphism, then $\mathrm{ChartableR2}(x) \iff \mathrm{ChartableR2}(h(x))$.

4.3 The Generic Boundary-Contrast Obstruction

Theorem 4.4 ([Novel] `ChartableR2Bridge.lean`).
(`not_homeomorphic_of_chartableR2_boundary_contrast`)
Let X and Y be topological spaces with predicates $b_X \subseteq X$ and $b_Y \subseteq Y$ such that:

(i) $\forall x,\ x \in b_X \iff \neg\,\mathrm{ChartableR2}(x)$,

(ii) $\forall y,\ y \in b_Y \iff \neg\,\mathrm{ChartableR2}(y)$,

(iii) $\mathrm{IsEmpty}(\mathrm{Subtype}(b_X) \simeq_t \mathrm{Subtype}(b_Y))$.

Then $\mathrm{IsEmpty}(X \simeq_t Y)$.

Proof sketch. Any homeomorphism $e : X \simeq_t Y$ preserves ChartableR2 (Theorem 4.3), hence maps b_X to b_Y pointwise. By e.subtype, we get a homeomorphism $\mathrm{Subtype}(b_X) \simeq_t \mathrm{Subtype}(b_Y)$, contradicting (iii). □

Theorem 4.4 subsumes the non-homeomorphism strategy of §6: any other pair of spaces satisfying the same three hypotheses is also not homeomorphic.

4.4 Half-Space Obstructions

Theorem 4.5 (`HalfPlaneVsPlane.lean`).
(`euclideanHalfSpace2_not_homeomorphic_euclidean2`)
$\mathrm{IsEmpty}(H^2 \simeq_t \mathbb{R}^2)$. *Proof via punctured half-plane (star-convex $\Rightarrow$ simply connected) vs. punctured plane (not simply connected, from* exp *covering).*

Theorem 4.6 (`HalfSpaceNeighborVsPlane.lean`).
(`isEmpty_homeomorph_subtype_openH2_zero_nbhd_subtype_openPlane`)
For open $V \subseteq H^2$ with $0 \in V$ and open $O \subseteq \mathbb{R}^2$, there is no homeomorphism $V \cong_t O$.

4.5 Cylinder C4: Boundary Biconditional

Theorem 4.7 (`CylinderChartableBoundary.lean`).
(`closedCylinder_boundaryUnion_iff_not_chartableR2`)
For all $x \in$ ClosedCylinder*:*

$$x \in \partial_{\mathrm{Cyl}} \iff \neg\,\mathrm{ChartableR2}(x).$$

5 Interior Charts on the Möbius Strip: The Seam Problem

5.1 Why the Seam Is Hard

Interior-height points of the fundamental domain fall into three regimes: strict interior $0 < x < 1$, where the quotient is injective on small boxes; the vertical seam $x \in \{0, 1\}$ with $t \neq \frac{1}{2}$, where trigonometric coordinates can separate fibers when the height window avoids the equator; and the seam at $t = \frac{1}{2}$, where $(0, \frac{1}{2}) \sim (1, \frac{1}{2})$ and a trigonometric chart degenerates. ChartableR2 at the equator quotient class is obtained by a piecewise Euclidean unfolding map (see below).

5.2 Open-Cell Charts

Theorem 5.1 (`MobiusChartableBoundary.lean`).
(`chartableR2_mobiusQuotientMk_of_Ioo_square`)
If $p \in (0, 1)^2$ *(strict interior of the fundamental domain), then* ChartableR2($[\![p]\!]$).

5.3 Seam Trig Charts (Away from Equator)

Definition 5.2 (`MobiusSeamChart.lean`, `MobiusSeamChartable.lean`). *The* saturated seam patch $S(t_0, \varepsilon, \delta)$ *is the union of the left and right* ε*-thickened seam strips at height window* $(t_0 - \delta, t_0 + \delta)$*, saturated under* $\sim$*.*

Theorem 5.3 (`MobiusSeamTrigInject.lean`).
(`onPatch_mobiusFDTrigCoord_eq_iff_mobiusRel0`)
When $\delta < |t_0 - \frac{1}{2}|$*, the trigonometric FD coordinate map* mobiusFDTrigCoord *satisfies:*
mobiusFDTrigCoord(p) = mobiusFDTrigCoord(q) *on the saturated patch if and only if* $p \sim q$*.*

Theorem 5.4 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_mem_mobiusSeamSaturatedPatch`)
For $t_0 \neq \frac{1}{2}$ *and appropriate* ε, δ*, every point*
in $\pi(S(t_0, \varepsilon, \delta))$ *is* ChartableR2*.*

5.4 The Equator Chart: Piecewise Unfolding

Definition 5.5 (`MobiusSeamChartableR2.lean`).
The fundamental-domain unfolding map
unfoldFD : $[0, 1]^2 \to \mathbb{R}^2$ *is:*

$$\mathrm{unfoldFD}(x, t) := \begin{cases} \mathrm{mobiusPlaneCoord}(x, t) & \text{if } x \leq \frac{1}{2}, \\ \mathrm{mobiusPlaneTrigSeamPartnerSub} & \text{if } x > \frac{1}{2}. \\ \quad \big(\mathrm{mobiusPlaneCoord}(x, t)\big) & \end{cases}$$

Lemma 5.6 (`MobiusSeamChartableR2.lean`). unfoldFD *is* $\sim$*-invariant: if* $p \sim q$ *then* $\mathrm{unfoldFD}(p) = \mathrm{unfoldFD}(q)$.

Theorem 5.7 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_left_seam_half`)
$\mathrm{ChartableR2}([\![(0, \frac{1}{2})]\!])$.

5.5 Assembly: All Interior Points Are ChartableR2

Theorem 5.8 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_interior_height`)
For every $p \in [0,1]^2$ *with* $0 < p_2 < 1$*:*

$$\mathrm{ChartableR2}([\![p]\!]).$$

5.6 Möbius C4: Boundary Biconditional

Theorem 5.9 (`MobiusChartableBoundary.lean`).
(`mobiusStripBoundary_iff_not_chartableR2_of_forall_off_edge_chartable`)
For all $x \in \mathrm{MobiusStrip}$*:*

$$x \in \partial_{\mathrm{Mob}} \iff \neg\, \mathrm{ChartableR2}(x).$$

6 M-FINAL: The Möbius Strip Is Not Homeomorphic to the Closed Cylinder

Theorem 6.1 (`MobiusSeamChartableR2.lean`, `ChartableR2Bridge.lean`).
(`mobiusStrip_not_homeomorphic_closedCylinder`)

$$\mathrm{IsEmpty}(\mathrm{MobiusStrip} \simeq_t \mathrm{ClosedCylinder})\,.$$

Proof sketch. Instantiate Theorem 4.4 with $b_X = \partial_{\mathrm{Mob}}$, $b_Y = \partial_{\mathrm{Cyl}}$, the biconditionals (Theorems 4.7 and 5.9), and the boundary incompatibility (Theorem 3.5). □

7 The Extended Master Theorem and Instantiations

7.1 Formal Statement

Theorem 7.1 (`Main.lean`).
(`representational_regress_master_extended`)
For any representational system R*:*

$$\mathrm{RepresentationalRegressMasterExtended}(R)$$

where this abbreviation expands to:

(i) $\mathrm{representational_regress_master_claim}(R)$
(regress + wall + T_2 *transport),*

(ii) $\mathrm{IsEmpty}(H^1 \simeq_t \mathbb{R}^1)$ *(1D half-space vs. line;*
`representational_regress_topology_halfLineModel`),

(iii) $\mathrm{IsEmpty}(\mathrm{MobiusStrip} \simeq_t \mathrm{ClosedCylinder})$ *(M-FINAL).*

This *extends* `representational_regress_master`; it does not replace it.

7.2 Bundled boundary models

Definition 7.2 (`ChartableR2Neighbor.lean`)**.** *A* `ChartableR2` boundary model *on X is a subset $b_X \subseteq X$ together with a proof that $\forall x,\ x \in b_X \iff \neg\,\mathrm{ChartableR2}(x)$.*

- The type `ChartableR2BoundaryModel` packages such data; the lemma `ChartableR2BoundaryModel.not_homeomorphic` restates Theorem 4.4 in bundled form.
- The development `ChartableR2ConcreteBoundaryModels` supplies `mobiusStripChartableR2BoundaryModel` and `closedCylinderChartableR2BoundaryModel`.
- The same-file theorem `mobiusStrip_closedCylinder_boundary_models_not_homeomorphic` re-proves M-FINAL along that pipeline.

7.3 Instantiations

Theorems earlier in the paper are unconditional. This subsection records how the same statements bear on four debates outside pure mathematics. Where an extra premise is required, it is stated explicitly; nothing in the formalization defaults a controversial premise to true.

7.3.1 Consciousness and self-presence

Philosophical bridge (premise P2). Some accounts distinguish representational self-reference from a stronger notion of *self-presence* (first-person givenness, non-representational acquaintance, etc.). Premise **P2** is *not* a theorem in the library: it says only that, for the purposes of a topological comparison, one class of phenomena is modeled by a space globally akin to MobiusStrip (here: one connected boundary band in the witness pair, quotient non-orientability in the usual informal picture) and another class by a space akin to ClosedCylinder (two disjoint boundary components, "two-sided" organization in the model).

What is proved. Theorem 6.1 is unconditional: MobiusStrip $\not\simeq_t$ ClosedCylinder. The proof localizes "boundary" via `ChartableR2` and compares boundary subspaces. Any other pair of compact spaces that admit the same pattern of biconditionals and subtype incompatibility is also not homeomorphic, by Theorem 4.4.

Conditional consequence. If (i) one treats iterative representation as living on a substrate whose `ChartableR2`-detected boundary matches the *cylinder* side of the development, and (ii) one accepts P2 for a target phenomenon modeled on the *Möbius* side, then no global homeomorphism can identify that substrate with the target: the obstruction is topological. Under those assumptions, resource limits (compute, context length, training data) address a different question; they do not alter homeomorphism type. Whether P2 is acceptable is a separate philosophical issue; the formal work neither proves nor assumes it.

7.3.2 Symbol grounding and semantic regress

In the symbol-grounding problem, arbitrary symbols must be interpreted, and interpreting the interpretation threatens a regress unless a non-representational stop is posited. Formally, a representational system already contains the morphism `represent` whose iterates index "one more representational step." The type alias `SymbolSystem` is definitionally identical to `Representational`

`System`; lemmas such as `symbolGrounding_metaRegressArrow_injective` record that distinct natural-number depths correspond to *distinct* endomorphisms of the awareness object. Thus semantic "levels of interpretation" inherit the same non-collapse facts as the categorical tower—not because the formalism defines "meaning," but because the regress of *structured* interpretation steps is what `metaRegressArrow` tracks.

7.3.3 Diagonal arguments and self-interpretation

Lawvere's theorem in `Type` implies that there is no family $s : A \to A \to B$ that realizes every unary $g : A \to B$ as $s(a)$ for some parameter a, once B carries an endomorphism without fixed points (e.g. `no_universal_parametric_unary_nat` for $B = \mathbb{N}$, or `no_universal_parametric_unary_bool` for $B = \mathrm{Bool}$ with $\neg$—the decidable/computability analogue).

Crucially, the diagonal obstruction does not require the adversary's self-model to be surjective. `lawvere_diagonal_not_in_range` shows that for *any* $s : A \to A \to B$ and any fixed-point-free $f : B \to B$, the diagonal function $a \mapsto f(s\,a\,a)$ is *never* equal to $s\,a_0$ for any a_0. In other words, even a *partial* self-interpreter necessarily has a blind spot containing its own diagonal—the adversary cannot circumvent the obstruction by abandoning totality.

`LawvereRegressBridge` collects such statements and connects them to the categorical presentation: in particular, `lawvere_fixed_point_stays_representational` says that for every n and every object witness c, the iterate $\mathrm{OntologicalSlot.endo}(\mathrm{represent}^n)$ is unequal to $\mathrm{OntologicalSlot.obj}(c)$. So however high the tower of self-applicative coding rises inside the type discipline, the formalism still distinguishes the morphism-slot tag from the global-element tag—a different statement from Lawvere's pointwise fixed-point existence for $B \to B$, but the intended hinge between diagonal phenomena and the `OntologicalSlot` reading of the wall.

7.3.4 Quantum measurement and observer chains

Talk of apparatus, environment, and Wigner's-friend hierarchies suggests a chain of systems, each described relative to a larger "observer." If one ever gives such a chain a *topological* state space Obs, together with a subset ∂_{obs} such that membership agrees with $\neg\,\mathrm{ChartableR2}$ as in the cylinder/Möbius story, and if one shows that ∂_{obs} is not homeomorphic to mobiusStripBoundary as a subspace, then Theorem 4.4 applies immediately: `QuantumObserverChainHypothesis` is only this data packaged as assumptions, and `QuantumObserverChainHypothesis.not_homeomorphic_mobiusStrip` is the conclusion $\mathrm{IsEmpty}(\mathrm{Obs} \simeq_t \mathrm{MobiusStrip})$. The present library does *not* define Hilbert spaces, observables, or projection postulates; any use of the hypothesis structure in physics requires an independent argument that the modeled readout topology satisfies its fields.

7.4 Scaling of language models and similar systems

Training larger models extends a representational architecture; it does not, by itself, produce a homeomorphism between a given topological model of that architecture and a different global model. If the relevant state space is modeled with cylinder-type boundary data and a target phenomenon demands Möbius-type data, Theorem 4.4 encodes the sense in which the gap is not closed by parameter count or depth alone.

7.5 Relationship to the Trans-Computational Processing argument

An earlier line of work argues from computability—halting, Gödel-style diagonalization, and related limits—to constraints on "self-contained" computational mind models. The topological strand here

is independent: it turns on `ChartableR2` sensors and boundary subspaces, not on recursiveness. The two approaches can reinforce a common *limitative* conclusion about representation closed under its own rules, but neither formally entails the other.

7.6 No-escape analysis: horn-by-horn closure

An adversary confronting the representational regress has, at first glance, several avenues of retreat. Each is formalized and shown to fail. The numbering below follows the order in which the escape was identified; each "horn" names an adversarial move, and the closing theorem blocks it.

Horn 1: Infinite injective tower. *Move:* accept iter_injective and try to use the tower itself as a self-representation resource. *Closure:* Theorem 2.2 yields infinitely many distinct meta-levels; Theorem 2.7 (the Lawvere wall) ensures the endo/obj distinction persists at every level. The tower does not collapse into closure.

Horn 2: Finite-order / periodic iterates. *Move:* weaken iter_injective—let $\text{represent}^n = \text{represent}^m$ for some $n \neq m$, so the loop has only finitely many distinct powers.
Closure: `IterateClosure.End.pow_iterate_dichotomy` proves the dichotomy: either the iterates are all distinct (Horn 1) or there exists a bound B such that every power equals some e^i with $i < B$. In the finite case the system's representational depth is bounded; it cannot host an injective tower at all.

Horn 3: Cross-object readout via section–retraction. *Move:* represent A *inside* a different object B using a section $s : A \to B$ and retraction $r : B \to A$ with $s \circ r = \mathrm{id}_A$. *Closure:* `CrossObjectRepresentation.section_retraction_no_injective_tower_either_side` shows that $r \circ s$ is idempotent on B (so $(r \circ s)^1 = (r \circ s)^2$, blocking injective powers on the host side) and $s \circ r = \mathrm{id}_A$ (so all powers are the identity on the source side—trivially periodic). Neither side supports an injective $\mathbb{N}$-tower.

Horn 4: Partial (non-surjective) self-model. *Move:* concede that no *total* self-interpreter exists, but claim a useful *partial* one. *Closure:* `lawvere_diagonal_not_in_range` requires no surjectivity of s: for *any* $s : A \to A \to B$ and any fixed-point-free $f : B \to B$, the diagonal function $a \mapsto f(s\,a\,a)$ is never in the range of s. Every partial self-model necessarily has a blind spot that includes its own diagonal—the information that matters most for self-capture is exactly the information it cannot represent.

Horn 5: Different topological models. *Move:* reject the cylinder/Möbius witness; use a different geometric model. *Closure:* Theorem 4.4 (`not_homeomorphic_of_chartableR2_boundary_contrast`) is *model-agnostic*: it applies to *any* pair of spaces whose ChartableR2-co-characterized boundaries are subtype-incompatible. The cylinder and Möbius strip are a discharged witness; any other pair meeting the same three hypotheses imports the same conclusion.

Horn 6: Computability / decidable self-reference. *Move:* restrict to decidable predicates or finite-valued functions, where "self-reference" seems more tractable than Cantorian diagonals. *Closure:* `no_universal_parametric_unary_bool` shows that no $s : A \to A \to \mathrm{Bool}$ can realize every $g : A \to \mathrm{Bool}$, since $\neg : \mathrm{Bool} \to \mathrm{Bool}$ is fixed-point-free. The obstruction is not an artifact of infinite cardinality; it holds in the decidable, two-valued setting.

Residual openings (not closed here). Two further routes are not formally addressed in this paper: *(i) higher-categorical self-maps* (a self-representation living in a 2-category or ∞-category, where the monoid of endomorphisms is replaced by a richer hom-object); and *(ii) non-categorical frameworks* (a system that resists categorical encoding entirely). The first is subject to analogous monoid-element arguments once a truncation to the 1-categorical endomorphism monoid is performed; the second falls outside the scope of any categorical formalization by definition. Both are noted as limitations in §8.

7.7 Admissibility: what the closures tell neighboring debates

Given that Horns 1–6 are closed, the formal landscape constrains what *admissible* solutions look like in each neighboring debate. The reasoning in every case is the same: the closures establish a structural barrier, so any proposed solution must either (a) operate outside the barrier's hypotheses, or (b) accept the consequences.

Consciousness and self-presence. If a system's state-space has cylinder-type boundary organization (two disjoint boundary components, each ¬ChartableR2), then no continuous global identification (homeomorphism) maps it to a Möbius-type target (one connected boundary band). An admissible account of self-presence must therefore either (i) posit that the relevant substrate *already* has the target topological type (in which case it was never purely representational in the cylinder sense), or (ii) invoke a discontinuous or non-topological bridge (which the formalism does not model and which would require independent justification). Scaling—more parameters, longer context windows, deeper architectures—extends a representational system *within* a fixed topological type; it does not change boundary organization.

Symbol grounding. Any symbol system that iterates interpretation is caught in the Horns 1/2 dichotomy: either infinite distinct levels (full regress plus Lawvere wall) or finite-bounded depth (finite loop, no injective tower). Horn 4 adds that even a *partial* interpretation function misses its own diagonal. Admissible grounding must therefore come from *outside* the representational loop: the system cannot ground itself by iterating its own interpretive machinery, whether totally or partially.

Diagonal arguments / AI self-interpretation. Any proposed self-interpreter $s : A \to A \to B$ (total or partial) necessarily fails to include its own diagonal (`lawvere_diagonal_not_in_range`); this holds for $B = \mathbb{N}$ (numeric outputs), $B = \mathrm{Bool}$ (decidable predicates), and any B with a fixed-point-free endomorphism. Admissible self-interpreters *must* be incomplete in a precise sense: they cannot model their own diagonal behavior. This is not a resource limitation but a structural one; no amount of compute, training data, or architectural ingenuity changes the diagonal exclusion.

Quantum measurement chains. If the measurement chain's state space has a boundary pattern matching the cylinder side of the development, then no homeomorphism identifies it with a Möbius-type target. The chain must either introduce genuinely new topological structure at some level—not representational iteration of the same type—or accept that its boundary organization differs from the target pattern.

LLM and AI scaling. Scaling extends a representational architecture within a fixed topological type. If that type has cylinder-side boundary organization, scaling does not produce a homeomorphism to a Möbius-type target. Admissible paths to a different topological type require *structural*

change (different topology of the underlying state space), not quantitative extension (more of the same).

8 Limitations and scope

1. **Hypothesis, not derivation.** iter_injective is bundled, not proved from minimal data. Not every $f : A \to A$ in every category satisfies it. The regress is conditional on this non-triviality hypothesis.

2. **Lawvere in `Type`, not abstract CCC.** The formalization uses Type u and MonoidalClosed(Type u), not a general closed category over arbitrary C. A fully abstract CCC version remains future work.

3. `OntologicalSlot` **is definitional, not metaphysical.** The disjointness endo $\neq$ obj is trivially true by constructor discrimination. This is the point, but the triviality means the result is *semantic* (about the structure of type theory) rather than a deep theorem.

4. **Models are chosen, not unique.** The closed cylinder and Möbius strip are *specific* topological models chosen for clarity and tractability. A different philosophical account might use different geometric witnesses (e.g., Klein bottle, different quotient). The argument works for any two models separated by a topological invariant; these two are the cleanest.

5. **P2 is philosophical.** The premise "consciousness requires non-orientable topology" is not a theorem. Section 7.3 states only what follows when P2 is adjoined to the proved non-homeomorphism; accepting or rejecting P2 remains an independent issue.

6. **No** IsManifold **polish.** The Möbius strip is not given a full Mathlib IsManifold instance. The proof uses ChartableR2 (open neighborhoods homeomorphic to $\mathbb{R}^2$) as a self-contained predicate, sidestepping the need for a full manifold-with-boundary atlas. This is sufficient for the non-homeomorphism but less elegant than a full atlas approach.

7. **Route W is incomplete.** An alternative proof via circle-winding / doubling obstructions (Steps 1–4 in `CylinderMobiusNonhomeo.lean`) is partially formalized. Step 5 (transport under a hypothetical global homeomorphism) is not proved. The flagship proof via ChartableR2 is complete and does not depend on Route W.

8. `QuantumObserverChainHypothesis`**.** This definition only bundles topological assumptions (boundary set, `ChartableR2` biconditional,
subtype non-homeomorphism to the Möbius boundary). It does not import Hilbert spaces, operators, or measurement axioms; a physical application would need a separate argument that a concrete model satisfies those fields.

9. **Partial self-models: blind spot, not total impossibility.** `lawvere_diagonal_not_in_range` proves that every partial self-model misses its own diagonal; it does not prove that partial self-models are *useless*—only that they are necessarily incomplete in a precise, diagonal-shaped way. Whether that incompleteness matters for a given application is a separate question.

10. **Higher-categorical and non-categorical frameworks.** The horn closures rely on 1-categorical endomorphism monoids. A self-representation living in a 2-category or ∞-category replaces the monoid with a richer hom-object; the present arguments apply after truncation

to the underlying 1-categorical monoid, but a native higher-categorical treatment is not given. Systems that resist categorical encoding entirely fall outside any categorical formalization by definition.

11. **Computability: decidable analogue only.** `no_universal_parametric_unary_bool` closes the Bool-valued Lawvere horn. A full treatment of Turing-degree obstructions (Rice's theorem, Kleene's recursion theorem) in the same framework is not attempted; the present result is the finite/decidable endpoint of the diagonal family.

9 Related work

9.1 Formalized topology

Mathlib's manifold and topology libraries underpin the half-space and chart lemmas; other proof assistants also host substantial topology. The weight here is on a concrete non-homeomorphism coupled to representational-system infrastructure, not on novel classification of differential invariants.

9.2 Lawvere's theorem in proof theory

Lawvere's fixed-point argument appears in multiple formal developments. `OntologicalSlot` and the wall lemmas supply complementary category-internal tagging alongside the `Type`-level diagonal theorem.

9.3 Consciousness, Gödel, and AI

Gödel- and computability-based limits on mechanism (including Penrose-style arguments, discussions of strange loops and self-reference in the sense of Hofstadter, and information-theoretic theories of consciousness) occupy a large literature orthogonal to the cylinder/Möbius homeomorphism question, though informal exposition sometimes elides the difference.

9.4 Topology in philosophy of mind

Topological figures of speech appear in some philosophy of mind. The present work differs in that its central topological claims are proved and the generalization pattern is explicit.

10 Conclusion

The formalization delivers three interlocking layers:

(i) A machine-checked regress and Lawvere infrastructure (categorical strand),

(ii) A general ChartableR2 boundary-contrast lemma with a fully discharged compact witness MobiusStrip $\not\simeq_t$ ClosedCylinder (topological strand), and

(iii) A systematic no-escape analysis (Horns 1–6, §7.6) that formally closes the principal adversarial retreat routes: infinite regress, finite-order collapse, cross-object readout, partial self-models, model substitution, and decidable self-reference.

The central insight is not that infinite regress is bad—that is widely acknowledged. The insight is that the barrier is **topological and structural**, not computational or resource-bound. The obstacle is *wrong type*, not *inexhaustible queue*. A homeomorphism obstruction is a finite-data invariant: two spaces are either homeomorphic or not, regardless of how many modeling steps one enumerates. No amount of compute, no deeper self-model, no richer architecture changes the boundary pattern of a topological type. And the Lawvere diagonal exclusion (`lawvere_diagonal_not_in_range`) shows that even *partial* self-models necessarily have a blind spot containing their own diagonal—the very information needed for self-capture is the information they cannot represent.

What is not being claimed. The theorems are unconditional mathematics; the *application* to any particular domain requires domain-specific premises (P2 for consciousness, `ChartableR2` biconditionals for a specific state-space model, etc.). The formalism does not prove that consciousness is non-representational, that AI cannot be conscious, or that symbol grounding is impossible. It proves that *if* the relevant state space has the boundary pattern characterized here, *then* a homeomorphism to a space with a different boundary pattern does not exist—and that the standard adversarial attempts to work around that barrier fail in formally checkable ways.

The admissibility picture. The horn closures (§7.6) together with the admissibility analysis (§7.7) leave a clean landscape. In each neighboring debate—consciousness, symbol grounding, diagonal limitations, measurement chains, AI scaling—the admissible solutions share a common shape: *structural change* to the topological type of the substrate, not *quantitative extension* within a fixed type. The formalism makes this precise and machine-checkable, so that future claims about circumventing the barrier can be evaluated against a fixed standard.

Future work. A full IsManifold instance for MobiusStrip; completion of the optional Route W winding track; a more abstract Lawvere statement for general cartesian closed categories; native higher-categorical horn closures (2-category and ∞-category analogues); further models where ChartableR2 boundary data can be established; and a full Turing-degree analogue of the decidable Lawvere horn.

A Lean Module Map

B Key Theorem Index

C Reproduction

```
cd representational-regress-lean
lake update
lake exe cache get
lake build RepresentationalRegress
# Verify: 0 sorry
rg "sorry" RepresentationalRegress/
```

Toolchain: `leanprover/lean4:v4.29.0-rc6`. Mathlib: `v4.29.0-rc6`.

Module	Role
`Basic`	`RepresentationalSystem`, iterates, `OntologicalSlot`
`Regress`	Non-termination, unbounded levels
`FixedPoints`	`EndoVsPoint`, Lawvere wall non-dissolution
`Lawvere`	Fixed-point theorem in `Type`
`LawvereCCCType`	`MonoidalClosed` packaging
`Orientability`	T_2 homeomorphism invariance
`CylinderMobius`	Fundamental domain, quotient, compact Hausdorff, boundary band
`CylinderBoundary`	Closed cylinder boundary: two disjoint connected faces
`CylinderChartable Boundary`	Cylinder C4 biconditional
`ChartableR2Neighbor`	`ChartableR2`, `ChartableR2BoundaryModel`; openness / stability
`ChartableR2Bridge`	`Homeomorph.chartableR2_iff`; `not_homeomorphic_of_chartableR2_boundary_contrast`; `ChartableR2BoundaryModel.not_homeomorphic`
`ChartableR2Concrete`	Möbius/cylinder boundary models; conditional observer chain; `Quantum`
`BoundaryModels`	`ObserverChainHypothesis`
`SymbolGrounding`	`SymbolSystem`; `symbolGrounding_*`
`LawvereRegressBridge`	`no_universal_parametric_unary_nat`, `no_universal_parametric_unary_bool`; `lawvere_diagonal_not_in_range`; `lawvere_fixed_point_stays_representational`
`NoEscapeClosure`	Iterate collisions vs. injective towers; section–retraction idempotent readout
`ChartableR2Models`	Interior charts; cylinder interior `ChartableR2`
`MobiusChartable Boundary`	Open-cell charts; boundary ¬`ChartableR2`; Möbius C4 packaging
`MobiusSeamChartable`	Saturated seam patches (open, saturated)
`MobiusSeamChart`	IFT-based local model; trig coordinates
`MobiusSeamTrigInject`	Trig injectivity on quotient windows
`MobiusSeamChartableR2`	Seam charts (trig + unfold); interior-height; **M-FINAL**
`MobiusCylinder BoundaryContrast`	Boundary subspaces not homeomorphic
`CylinderMobius Nonhomeo`	Route W (optional; Steps 1–4)
`HalfLineVsLine`	1D half-space $\neq$ line
`HalfPlaneVsPlane`	2D half-space $\neq$ plane
`HalfSpaceNeighborVs Plane`	Local half-space $\neq$ open plane patch
`PuncturedPlaneNot SimplyConnected`	$\mathbb{C} \setminus \{0\}$ not simply connected
`OpenCompactChart Obstruction`	Compact $\mathbb{R}^2$ patch $\neq \mathbb{R}^2$
`Main`	Master theorem + extended bundle

Lean name	Statement	Module
regress_iterates_unbounded	$\forall B, \exists L > B$, distinct iterate	Regress
lawvere_fixed_point_Type	Universal enumerator $\Rightarrow$ every endo has fixed point	Lawvere
lawvere_wall_is_not_dissolution	endo(represent) $\neq$ obj(fp)	FixedPoints
instT2SpaceMobiusStrip	Möbius strip is T_2	CylinderMobius
isConnected_mobiusStripBoundary	Boundary band is connected	CylinderMobius
closedCylinder_boundary_eq_union	Cylinder boundary = 2 disjoint faces	CylinderBoundary
euclideanHalfSpace2_not_homeomorphic_euclidean2	$H^2 \not\cong_t \mathbb{R}^2$	HalfPlaneVsPlane
not_homeomorphic_of_chartableR2_boundary_contrast	Generic: incompatible boundaries $\Rightarrow \neg$ homeo	ChartableR2Bridge
closedCylinder_boundaryUnion_iff_not_chartableR2	Cylinder C4 biconditional	CylinderChartableBoundary
not_chartableR2_of_mem_mobiusStripBoundary	Möbius boundary $\Rightarrow \neg$ChartableR2	MobiusChartableBdy
chartableR2_mobiusQuotientMk_of_interior_height	Interior height $\Rightarrow$ ChartableR2	MobiusSeamChartableR2
mobiusStripBoundary_not_homeomorphic...	Boundary subtypes not homeo	MobiusCylBdyContrast
mobiusStrip_not_homeomorphic_closedCylinder	**M-FINAL**	MobiusSeamChartableR2
representational_regress_master	Core master conjunction	Main
representational_regress_master_extended	Extended bundle (+ topology)	Main
QuantumObserverChainHypothesis.not_homeomorphic_mobiusStrip	Conditional: obs. chain $\not\cong_t$ Möbius	ChartableR2ConcreteBoundaryModels
no_universal_parametric_unary_nat	No universal $s : A \to A \to \mathbb{N}$ for all $g : A \to \mathbb{N}$	LawvereRegressBridge
IterateClosure.End.pow_iterate_dichotomy	Injective iterate powers vs. finite index bound	NoEscapeClosure
CrossObjectRepresentation.section_retraction_no_injective_tower_either_side	Both-side: $\neg$ injective $\mathbb{N}$-powers on source *or* host	NoEscapeClosure
lawvere_diagonal_not_in_range	Diagonal never in range of any $s : A \to A \to B$ (no surjectivity)	LawvereRegressBridge
no_universal_parametric_unary_bool	No universal $s : A \to A \to \mathrm{Bool}$ for all $g : A \to \mathrm{Bool}$	LawvereRegressBridge